Electric shaRed mOBility hUbs Trial (ROBUST): Waterford 7-8 October 2026 workshops follow-up

James Reynolds

9 October, 2026

Dear participants of the Waterford co-design workshops[1],

Thank you for your inputs at the co-design workshops this week. There was a lot to take in from all of your comments. We will be looking closely at the materials developed in the workshop over the next weeks and months, with an aim to using your insights to improve the trial and better understand how shared electric mobility hubs might succeed here in Ireland or elsewhere.

[1] 6 and 7 October

Initial takeaways

My initial takeaways from your comments are:

- there are challenges in using the apps and other aspects of the user interface[2];
- reliability and availability may be limiting the potential of the hub[3], but might be improved on with more local and community involvement in the day-to-day management[4];
- topography is an important factor for the use of the e-bikes, given the location of the hub at the top of a hill[5]
- there may be potential to provide weather protected and well sheltered seating, as there is not much other than the bandstand nearby (and that just has steps).
- there may be potential for a bit of friendly competition between Waterford and the other hubs as far as tracking a leaderboard of 'most trips', 'most carbon saved' or something similar;
- there may be potential for alternate vehicle types, given the demographics (i.e. older population)[6] and other characteristics of the surrounding area (i.e. more apartment living and interest in car-free life[7], common use of walking and cycling[8]).

[2] Including for *new users*, for whom separate step-by-step instructions for the e-cars and the e-bikes in multiple formats (especially on-site signage) might help. However, the need to *both plug in and ALSO activate* the car charging with the access card is also an issue - not necessarily obvious to all users who might be expecting plugging in the cars would be enough to start recharging.

[3] Especially with respect to the cars' batteries running down when plugged in but *not* charging.

[4] Potential for local shops, community members or others to be involved in regularly monitoring the vehicles, taking them for a spin around the block every few days, acting as an immediate contact to help users etc. perhaps in exchange for a reduced rate, free hours, (in the case of shops) free use of cargo bikes for short deliveries / dropping off purchases just made in store, etc. .

[5] Need to walk up the hill to get to the hub, e-bikes not necessarily having the power or sufficient remaining charge to get up hills (when returning to the hub), braking while going downhill (especially for first-time riders).

[6] Electric tricycle? Tandem bike so someone can take someone else with them?

[7] Potential for e-cars, e-vans, additional e-cargobikes

[8] Many people already have their own bikes so shorter-range e-bikes may be less useful, but it might be useful to have longer range e-bikes for longer trips.

Immediate steps

Sticker for the car chargers

A sticker on the car charger to remind users to start the recharging (using the card inside the car) was suggested as a way to reduce problems with the cars being left plugged in but not 'charging'. I've put together a first version[9].

I'd suggest that all is needed is for someone who lives nearby to print out four of these on stickers[10], and attach one to each end of each of the charging cables.



I've tried it on a (different type of) car charger here in Sligo, and it looks ok as a concept. Maybe it might work on the ROBUST system in Waterford?

Pop-ups (seating?)

We touched on previous research findings about legitimising implementation through pragmatic strategies[11]

There may be opportunities in the short term to provide additional seating through bottom-up and incremental action. This might also be experimented with through adopting a pop-up approach to see what might work, what the community might like to have, or even if the community wants anything at all[12]. Watch out for that (possibly) happening sometime soon near the hub!

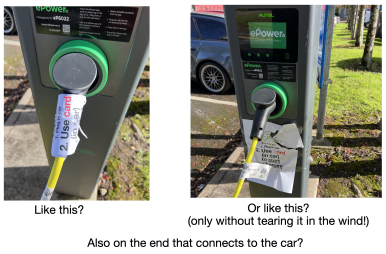


Figure 1: First version of sticker for car chargers

[9] available at [https://github.com/James-Reynolds/WP4_co-creative_workshops/blob/main/graphics/charger_sticker_261009.png] or [https://github.com/James-Reynolds/WP4_co-creative_workshops/blob/main/graphics/charger_sticker_half_size_261009.png]

Figure 2: First version of sticker for car chargers - test in Sligo on similar charger

[10] Or get the Design Print House to do it



Figure 3: QR code for short 2-pager with details about pop-ups and otherwise legitimising implementation

[11] See https://james-reynolds.github.io/implementation_transit_priority_handout.html (QR code above). The pragmatic strategies are: legitimation *before* implementation (1. Technical evaluation, 2. vision-based plans, 3. public enquiry/decision-making); avoiding impacts (4. grade-separation, 5. additional capacity, 6. subservience); and legitimation *through* implementation (7. bottom-up and incremental, 8. *pop-ups*, 9. formal trials).

[12] Pop-downs are always possible!

Information - online

As discussed in the workshops, we now have a draft short video on the project and how to use the e-bikes and e-cars. There's also an initial draft website. Please let us know if you have any initial thoughts, and stay tuned for further versions.



Draft intro video
https://www.youtube.com/watch?v=hu9jsgi8CVU



Draft mobility hub website
https://james-reynolds.github.io/ROBUST_Electric_shaRed_mOBility_hUbS_Trial_website/



Draft mobility hub website, Github source code
https://github.com/James-Reynolds/ROBUST_Electric_shaRed_mOBility_hUbS_Trial_website

Figure 4: Draft mobility hub video and website links

A great thing about co-design research is that *everyone* can take part in improving the output if they would like too! If you are interested in having a go and making your own videos or other forms of information about the hub please feel free to just go ahead. Drop us a line if you'd like to collaborate or get some help to get started. This is all (just) research and a trial to see what we can make, so nothing has to be perfect. . . [13]

Likewise, the website is hosted on Github (QR code and link above, right), and uses open source software. This means that you can edit it yourself and send me a 'pull request' to incorporate your changes. There are further details in the Readme of the github repository, or you can drop me an email or otherwise reach out if you'd like help getting started.

[13] . . . just hopefully better than what we've got so far - "A narrator with an Aussie accent? That's ridiculous! Can someone with an Irish accent send me a different version. Anyone who speaks Gaelic want to have a go at recording a version?"

Information - physical

Thanks for the feedback on the research work done to date on developing Mobility Hub informational and directional signs. The software developed so far is available on github[14] as well as PDF versions of the Waterford totem / pedestal signs that we had printed out in Ao size at the workshop[15]

Note of course, that these a rough outlines and just what we have so far. I'll be looking towards improving these based on your comments in the workshops[16], and iterating improved designs. If you like coding and are interest, you can (like the website) download

[14] https://github.com/James-Reynolds/mobilityHubTools

[15] See https://github.com/James-Reynolds/WP4_co-creative_workshops/tree/main/graphics_totems

[16] Especially adding information about the topography!

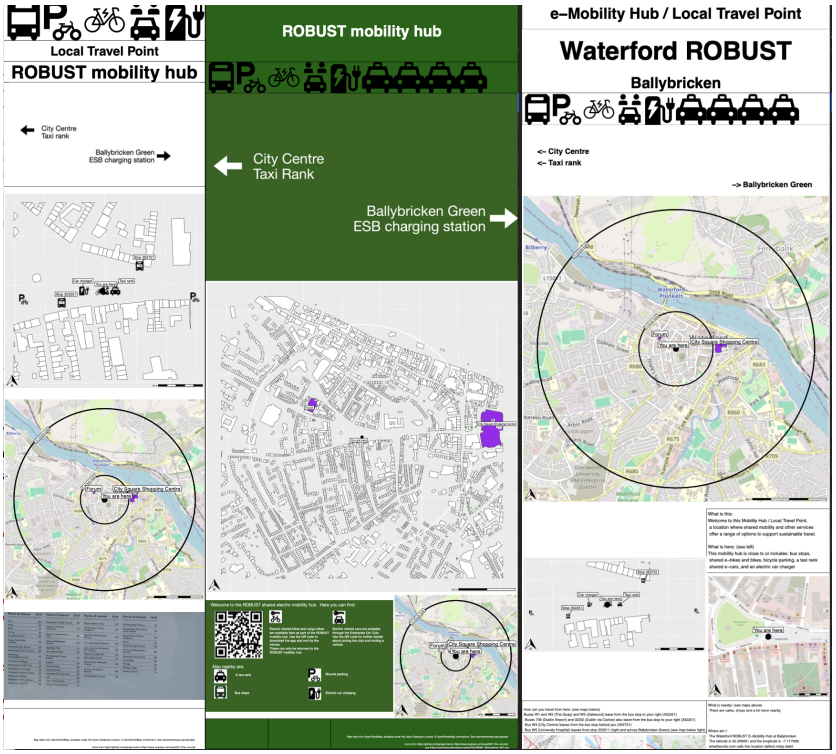


Figure 5: Waterford totems as per Bristol (left), Birmingham (centre) and a combined (right) style

the code, improve it, and send me a pull request to incorporate you improvements.

Please stay tuned for further details of what I come up with - perhaps we might be able to test some of them in the field and see what people on the street think?

More workshops?

Noreen and I are off to Letterkenny and Galway over the next month or so to run workshops there. But, neither of us is far away, and we'd certainly be interested in coming back to get your further thoughts an insights as the trial progresses and the iterative co-design processes iterate!

More generally, I'm happy to come talk or otherwise interact about transport, implementation and policy-making, or pretty much any other related topic, so please let me know if that is of interest.

All the best, and thanks for your inputs to the research

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